MIL-C-63318 (AR)
18 May 1979
SUPERSEDING
FA-PD-MI-2617
1 May 1964

MILITARY SPECIFICATION

CASE, CARTRIDGE, 165MM, M104
METAL PARTS ASSEMBLY

This specification is approved for use by the U.S. Army Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Case, Cartridge, 165MM, M104 Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-45467 - Cartridge, 165MM, HEP, M123A1, Loading, Assembling and Packing

FEDERAL

PPP-F-320 - Fiberboard, Sheet Stock and Cut Shapes
PPP-B-636 - Box, Fiberboard

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking of Inert Ammunition Components

DRAWINGS

ARRADCOM

8845066 - Case, Cartridge, 165MM, M104, Metal Parts Assembly
8845042 - Cartridge, 165MM, HEP, M123A1 Loading Assembling and Packing

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI - B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, New York 10018).

ASTM - E18 Tests for Rockwell Hardness and Rockwell Superficial Hardness for Metalic Materials
ASTM - B117 Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements of the applicable drawings and referenced specifications.

3.2 Heat treatment/stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall so be conducted so as to produce the mechanical properties and physical properties specified on the applicable drawing.

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3.3 Brazed joint.-The brazed metal shall be visible for the full 360° circumference around the exterior and interior of the joint.

3.3.1 Strength of brazed joint.-The body and case spring of the cartridge case shall be joined in the manner specified on drawing (dwg.) 8845066. The joint shall withstand the application of a 500 inch-pound, minimum, torque without permanent distortion or failure of the brazed joint.

3.4 Surface preparation and application of protective finish.-Where cleaning and surface pretreatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces designated by the applicable drawings. Salt spray and other required testing shall be in accordance with the applicable drawings and specifications.

3.5 Proving ground test.-There shall be no failure of the cartridge case assembly when tested as specified in 4.5.4.

3.6 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following.

3.7.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.7.2 Burr.-No parts shall have a burr.

3.7.3 Metal defects.-The cartridge case and components shall be free from cracks, splits, cold shuts, and inclusions, porosity or any similar characteristic.

3.7.4 Foreign matter.-The cartridge case and components shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE

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4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Case, Cartridge	8845066	35
Spring, Primer	8845070	10
Retainer, Spring	8845069	10
Case, Spring	8845068	10
Body	8845067	10
Screw	10520028	10
Handle Assembly	8845053	10
Retainer, Handle	8845058	10
Handle	8845057	10
Lock Assembly	8845059	10
Clip, Lock	8845061	10
Spring, Lock (Plated)	8845060	10
Spring, Lock (Unplated)	8845060	5

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Inspection Level II.

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4.4.2.1 Visual inspections.-In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ASA - B46.1 shall be used as a basis for surface roughness determination.

4.4.2.1.1 Defect reporting.-The contractor shall prepare a report of all critical defects found, including the number of cartridge case metal parts inspected and corrective action taken, in accordance with Data Item Description DI-R-1721, and deliver the report as specified in the contract.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Case, Cartridge, 165MM, M104 Metal Parts Assembly		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Cartridge (Dwg. 8845066)				
	Examination for defects	10		3.1	4.4.2.4
	Strength of brazed joint	10		3.3.1	4.5.1
	Body (Dwg. 8845067)				
	Examination for defects	10		3.1	4.4.2.2
	Case, Spring (Dwg. 8845068)				
	Examination for defects	10		3.1	4.4.2.3
	Lock Assembly (Dwg. 8845059)				
	Examination for defects	10		3.1	4.4.2.7
	Spring, Lock (Dwg. 8845060)				
	Examination for defects	10		3.1	4.4.2.5
	Determination of hardness	5		3.2	4.5.3
	Clip, Lock (Dwg. 8845061)				
	Examination for defects	10		3.1	4.4.2.6
	Handle Assembly (Dwg. 8845053)				
	Examination for defects	10		3.1	4.4.2.11
	Handle (Dwg. 8845057)				
	Examination for defects	10		3.1	4.4.2.8

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Retainer, Handle (Dwg. 8845058)				
	Examination for defects	10		3.1	4.4.2.9
	Retainer, Spring (Dwg. 8845069)				
	Examination for defects	10		3.1	4.4.2.12
	Spring, Primer (Dwg. 8845070)				
	Examination for defects	10		3.1	4.4.2.13
	Screw (Dwg. 10520028)				
	Examination for defects	10		3.1	4.4.2.10
	Ballistic Test	25		3.5	4.5.4

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Body		8845067
			NEXT HIGHER ASSEMBLY 8845066

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread, max.		0.40%	3.1	Gage
102	Minor diameter of thread, max.		0.40%	3.1	Gage
103	True position of body outside diameter at rear of body with pitch diameter of thread and front face		0.40%	3.1	Gage
104	True position of primer hole with pitch diameter of thread and front face		0.40%	3.1	Gage
105	True position of body outside diameter at front of body with pitch diameter of thread and front face		0.40%	3.1	Gage
106	Outside diameter of body at front of body		0.40%	3.1	Gage
107	Diameter of primer hole, max.		0.40%	3.1	Gage
108	Depth of primer hole recess		0.40%	3.1	Gage
109	Length of effective thread, min.		0.40%	3.1	Gage
110	Wall thickness at rear of body		0.40%	3.1	Gage
111	Body sidewall thickness		0.40%	3.1	Gage
112	Wall thickness at datum location, min.		0.40%	3.1	Gage
113	Angle of primer recess		0.40%	3.1	Gage
114	Wall thickness at primer hole recess		0.40%	3.1	Gage
115	Metal defective		100%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET				8845067
					NEXT HIGHER ASSEMBLY
					8845066
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pitch diameter of thread, min.		0.65%	3.1	Gage
202	Outside diameter of body at rear of body		0.65%	3.1	Gage
203	Diameter of primer hole, min.		0.65%	3.1	Gage
204	Diameter of holes in sidewall (2 sets)		0.65%	3.1	Gage
205	Diameter of holes in base		0.65%	3.1	Gage
206	Length, total		0.65%	3.1	Gage
207	Location of holes in sidewall (2 sets)		0.65%	3.1	Gage
208	Location of holes in base		0.65%	3.1	Gage
209	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
210	Surface finish improper		0.65%	3.1	Visual/Manual
211	Presence of burr		0.65%	3.7.2	Visual/Manual
212	Foreign matter present		0.65%	3.7.4	Visual
213	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Case, Spring		8845068
			NEXT HIGHER ASSEMBLY
			8845066

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of internal indents w/cavity		0.40%	3.1	Gage
102	True position of slots with cavity		0.40%	3.1	Gage
103	True position of indent with cavity		0.40%	3.1	Gage
104	Diameter of cavity, max.		0.40%	3.1	Gage
105	Length across internal indent, max.		0.40%	3.1	Gage
106	Length of sidewall indent		0.40%	3.1	Gage
107	Length from sidewall indent to end of lug		0.40%	3.1	Gage
Minor					
201	Diameter of flange		0.65%	3.1	Gage
202	Diameter of body		0.65%	3.1	Gage
203	Total length including slots		0.65%	3.1	Gage
204	Length between slots		0.65%	3.1	Gage
205	Width of indent		0.65%	3.1	Gage
206	Height of lugs		0.65%	3.1	Gage
207	Radius missing or incorrect		0.65%	3.1	Visual
208	Surface finish improper		0.65%	3.1	Visual/Manual
209	Presence of burrs		0.65%	3.7.2	Visual/Manual
210	Foreign matter		0.65%	3.7.4	Visual
211	Angle of flange improper		0.65%	3.1	Visual
212	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Case, Cartridge, 165MM, M104				8845066
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective			3.7.3	Visual
Major					
101	True position of mouth of spring case with pitch diameter of thread and open end face		0.40%	3.1	Gage
102	Depth to end of spring case		0.40%	3.1	Gage
103	Brazed joint inadequate		0.40%	3.3	Visual
104	Brazed joint torque test		100%	3.3.1	Test/4.5.1
Minor					
201	Marking, missing, incorrect or illegible		0.65%	3.1	Visual
202	Protective finish inadequate or missing		a	3.4	Test/4.5.2
203	Location of stamping		0.65%	3.1	Visual
204	Presence of burrs		0.65%	3.7.2	Visual/Manual
205	Foreign matter present		0.65%	3.7.4	Visual
206	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.5	Spring, Lock		8845060
			NEXT HIGHER ASSEMBLY
			8845059

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole, max.		0.40%	3.1	Gage
102	Length from top of angular bend to base		0.40%	3.1	Gage
103	Metal defective		100%	3.7.3	Visual
104	Hardness	10		3.1	Test/4.5.3
Minor					
201	Diameter of hole, min.		0.65%	3.1	Gage
202	Depth of slot		0.65%	3.1	Gage
203	Total length		0.65%	3.1	Gage
204	Length to angular bend		0.65%	3.1	Gage
205	Width of slot		0.65%	3.1	Gage
206	Thickness of spring		0.65%	3.1	Gage
207	Location of slot		0.65%	3.1	Gage
208	Location of hole		0.65%	3.1	Gage
209	Height of upright bend		0.65%	3.1	Gage
210	Width of spring		0.65%	3.1	Gage
211	Angle of bend incorrect		0.65%	3.1	Gage
212	Radius missing or incorrect		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET				8845060
					NEXT HIGHER ASSEMBLY
					8845059
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
213	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
214	Surface finish improper		0.65%	3.1	Visual/Manual
215	Presence of burrs		0.65%	3.7.2	Visual/Manual
216	Foreign matter present		0.65%	3.7.4	Visual
217	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8845061
4.4.2.6	Clip, Lock				NEXT HIGHER ASSEMBLY 8845059
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width, inside, max.		0.40%	3.1	Gage
102	Metal thickness, min.		0.40%	3.1	Gage
103	Metal defective		100%	3.7.3	Visual
Minor					
201	Depth, inside, to bend		0.65%	3.1	Gage
202	Length, outside, from base to bend		0.65%	3.1	Gage
203	Width, inside, min.		0.65%	3.1	Gage
204	Width across loop		0.65%	3.1	Gage
205	Metal thickness, max.		0.65%	3.1	Gage
206	Length of clip		0.65%	3.1	Gage
207	Radius missing or incorrect		0.65%	3.1	Visual
208	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
209	Surface finish improper		0.65%	3.1	Visual/Manual
210	Presence of burrs		0.65%	3.7.2	Visual/Manual
211	Foreign matter present		0.65%	3.7.4	Visual
212	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Lock Assembly				8845059
					NEXT HIGHER ASSEMBLY
					8845053
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from lock spring bend to base of lock clip, max.		0.40%	3.1	Gage
102	Improper assembly		0.40%	3.1	Visual
103	Any part missing		0.40%	3.1	Visual
Minor					
201	Crimp improper		0.65%	3.1	Visual
202	Presence of burrs		0.65%	3.7.2	Visual/Manual
203	Foreign matter present		0.65%	3.7.4	Visual/Manual
204	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Handle		8845057
			NEXT HIGHER ASSEMBLY
			8845053

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Metal defective		100%	3.7.3	Visual
Minor					
201	Length of opening between ends		0.65%	3.1	Gage
202	Length of pinched notches		0.65%	3.1	Gage
203	Length, inside, of handle		0.65%	3.1	Gage
204	Width across pinched notches		0.65%	3.1	Gage
205	Width of protrusion os pinched notches		0.65%	3.1	Gage
206	Handle thickness		0.65%	3.1	Gage
207	Location of pinched notches		0.65%	3.1	Gage
208	Radius or chamfer missing or incorrect		0.65%	3.1	Visual
209	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
210	Presence of burr		0.65%	3.7.2	Visual/Manual
211	Foreign matter present		0.65%	3.7.4	Visual
212	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.9	Retainer, Handle				8845058
					NEXT HIGHER ASSEMBLY
					8845053
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of holes in sidewall, max.		0.40%	3.1	Gage
102	Depth of spring slot, max.		0.40%	3.1	Gage
103	Width of spring slot, max.		0.40%	3.1	Gage
104	Thickness of wall		0.40%	3.1	Gage
105	Metal defective		100%	3.7.3	Visual
Minor					
201	Diameter of holes in sidewall, min.		0.65%	3.1	Gage
202	Diameter of cavity		0.65%	3.1	Gage
203	Diameter across lug end		0.65%	3.1	Gage
204	Diameter across open areas, lug end		0.65%	3.1	Gage
205	Diameter of hole opposite spring slot		0.65%	3.1	Gage
206	Depth of flat, min.		0.65%	3.1	Gage
207	Depth of spring slot, min.		0.65%	3.1	Gage
208	Length across opposite lugs		0.65%	3.1	Gage
209	Length across opposite bent lugs		0.65%	3.1	Gage
210	Length of bent lug		0.65%	3.1	Gage
211	Total length		0.65%	3.1	Gage
212	Width of flat		0.65%	3.1	Gage
213	Width of lugs		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.9	CONTINUATION SHEET				8845058
					NEXT HIGHER ASSEMBLY
					8845053
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
214	Width of spring slot, min.		0.65%	3.1	Gage
215	Location of holes in side		0.65%	3.1	Gage
216	Location of lugs		0.65%	3.1	Gage
217	Location of hole opposite spring slot		0.65%	3.1	Gage
218	Radii or angles missing or incorrect		0.65%	3.1	Visual
219	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
220	Surface finish improper		0.65%	3.1	Visual/Manual
221	Presence of burr		0.65%	3.7.2	Visual/Manual
222	Foreign matter present		0.65%	3.7.4	Visual
223	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Screw				10520028
					NEXT HIGHER ASSEMBLY
					8845066
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length of thread		0.65%	3.1	Gage
202	Thread damaged or not full		0.65%	3.7.1	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Handle Assembly				8845053
					NEXT HIGHER ASSEMBLY 8845066
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper assembly		0.40%	3.1	Visual
102	Any part missing		0.40%	3.1	Visual
Minor					
201	Total length, max.		0.65%	3.1	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Retainer, Spring				8845069
					NEXT HIGHER ASSEMBLY
					8845066
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of recesses with center hole		0.40%	3.1	Gage
102	True position of tabs with center hole		0.40%	3.1	Gage
103	Diameter across tabs, min.		0.40%	3.1	Gage
104	Diameter of center hole, max.		0.40%	3.1	Gage
105	Length of tabs		0.40%	3.1	Gage
106	Metal defective		100%	3.7.3	Visual
Minor					
201	Diameter across recess		0.65%	3.1	Gage
202	Diameter across tabs, max.		0.65%	3.1	Gage
203	Diameter of center hole, min.		0.65%	3.1	Gage
204	Thickness		0.65%	3.1	Gage
205	Length of recesses		0.65%	3.1	Gage
206	Surface finish improper		0.65%	3.1	Visual/Manual
207	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
208	Presence of burr		0.65%	3.7.2	Visual/Manual
209	Foreign matter present		0.65%	3.7.4	Visual
210	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Spring, Primer		8845070
			NEXT HIGHER ASSEMBLY
			8845066

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load at specified height		0.40%	3.1	Gage
102	Solid length, max.		0.40%	3.1	Gage
103	Number of active coils incorrect		0.40%	3.1	Visual/Manual
104	Number of total coils incorrect		0.40%	3.1	Visual/Manual
105	Direction of helix turn incorrect		0.40%	3.1	Visual
Minor					
201	Wire diameter, max.		0.65%	3.1	Gage
202	Outside diameter, max.		0.65%	3.1	Gage
203	Inside diameter, min.		0.65%	3.1	Gage
204	Diameter of loop at end		0.65%	3.1	Gage
205	Location of loop at end		0.65%	3.1	Gage
206	Profile of ends		0.65%	3.1	Gage
207	Protective finish inadequate or missing		a	3.4	Visual/4.5.2
208	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. See 4.4.3.1 for sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Cartons, Packed, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Support missing or damaged		0.65%	3.1	Visual
202	Separator(s) missing or damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63318 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Cartons, Packed, After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.1	Visual/Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not properly sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Surface preparation and application of protective finish.-The sample size and acceptance criteria for the inspection of surface preparation and the application of protective finish shall be in accordance with the applicable specifications. Failure to comply with the applicable specification and drawing requirements shall be cause for taking the corrective action recommended therein (see 4.5.2).

4.4.3.2 Hardness.-Ten spring locks from each lot shall be subjected to the hardness test. Failure of any spring lock to comply with the requirements of dwg. 8845060 shall be cause for rejection of the lot when tested as specified in paragraph 4.5.3.

4.4.3.3 Proving ground test.-Ten cartridge case assemblies from each lot shall be tested for safety in accordance with the procedures of paragraph 4.5.4. The results of the post firing and recovery examination shall be recorded on the firing record for information only. Failure of any cartridge case assembly for safety shall result in the rejection of the lot.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.2 herein and Section 6 of MIL-A-48078).

4.5 Test methods and procedures

4.5.1 Strength of brazed joint.-The cartridge case assembly shall be mounted in a fixture and held to prevent rotation and slippage. A torque wrench, using an appropriate adapter and set at 500 inch-pounds, minimum, shall be fitted to the spring case for the application of the required amount of torque. Observation shall be made for visual evidence of distortion or failure of the brazed joint.

4.5.2 Surface preparation and application of protective finish.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and acceptance criteria for testing as specified in the applicable specification. A salt spray (fog) test shall be performed on components in accordance with the requirements of the applicable drawings and specifications, using the equipment and test procedures prescribed by ASTM B117.

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4.5.3 Hardness.-The test shall be performed on the spring lock sample with a hardness tester prior to the application of the protective finish to determine compliance with the requirements of dwg. 8845060. The test shall be performed in accordance with the procedures prescribed for Rockwell Hardness Testing by ASTM E18.

4.5.4 Proving ground test.-The samples ballistically tested as prescribed herein shall be fired in a M135 gun. The ammunition shall be conditioned to $70^{o} \pm 2.5^{o}F$ ($21.1^{o} \pm 1.5^{o}C$) until all propellant is stabilized to a uniform temperature prior to firing, which normally requires a period of twenty-four hours, minimum. Maximum chamber pressure shall be recorded for all rounds (see 6.4). The propelling charge weight shall be adjusted to obtain a minimum mean pressure of 5600 psi (38.6 Mega Paschals (M Pa)) and a maximum individual pressure not to exceed 6000 psi (41.4 M Pa) for all samples fired in the proving ground tests.

4.5.4.1 Loading procedure.-The cartridge case metal parts assembly and components shall be assembled into cartridges in accordance with dwg. 8845042 and MIL-C-45467, except that an inert fill shall be substituted for the high-explosive filler and an inertly-loaded M62A2 fuze shall be used.

4.5.4.2 Camera coverage.-Adequate photographic equipment shall be used to assure satisfactory coverage of each sample's integrity and flight from the gun muzzle to the first one hundred feet of projectile flight, minimum.

4.5.4.3 Safety.-All samples fired in proving ground tests shall be observed for safety, and inparticular, the separation of the cartridge case from the projectile body prior to impact with the intended target. Any sample failing the safety criteria above shall be cause for the cessation of the remainder of the ballistic test in which the failure occurs and shall reject the lot or first article submission represented by the ballistic sample.

4.5.4.4 Examination of recovered case assemblies.-Following their firing and subsequent recovery, each cartridge case shall be examined for evidence of splits and separation from the projectile body. If a separation is found and is attributable to ground impact, the firing record will be so annotated. If a projectile (cartridge case) is not recoverable, its photographic record of firing (smear photo) shall be examined for evidence of splits or separation.

5. PREPARATION FOR DELIVERY

5.1 Packing

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5.1.1 Cartridge case assembly.-Fifteen (15) cartridge case assemblies shall be packed, Level B, in accordance with MIL-STD-1169 in a 3 X 5 pattern with a separator between each row or column within a Style RCS, Type CF, Class, Domestic, Variety SW, Grade 200 Box.

5.1.2 Retainer spring.-Five hundred (500) retainer springs shall be packed, Level B, in accordance with MIL-STD-1169, ten per cell in a 10 X 5 pattern within a Style RCS, Type CF, Class, Domestic, Variety SW, Grade 200 Box.

5.1.3 Handle assembly.-Fifty (50) handle assemblies shall be packed, Level B, in accordance with MIL-STD-1169 in 5 layers with a filler between layers in a 5 X 2 pattern within a Style RCS, Type CF, Class, Domestic, Variety SW, Grade 200 Box.

5.1.4 Primer spring.-Five hundred (500) primer springs shall be packed, Level B, in accordance with MIL-STD-1169 in 10 layers with a filler between each layer in a 10 X 5 pattern within a Style RCS, Type CF, Class, Domestic, Variety SW, Grade 200 Box.

5.2 Marking and labeling.-Marking and labeling of the cartons shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering Data

6.1.1 Procurement requirements.-See MIL-A-48078

6.1.2 Data requirements.-Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.2.1.1	Quality Inspection Defect Report	DI-R-1721

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

6.1.2.1 Delivery of data to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of one copy, each, to the technical agency, of the Ammunition Data Cards and Quality Inspection Defect Reports prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

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6.2 Submission of designs for approval.-Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801 (See 6.2.3 of MIL-A-48078).

6.3 Changes from previous issue.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue of this specification.

6.4 Firing record information.-The ballistic test firing records shall include the serial numbers of all weapon and fire control components. Stock component lot numbers of warmers and/or spotters shall also be included. For each round fired, the record shall contain deflection, elevation, time of firing, chamber pressure, pertinent flight characteristics, and the results of the post-recovery examination. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E and DRDAR-LCU-CV, Dover, New Jersey 07801.

6.4.1 Impact area.-During the recovery portion of the ballistic test, the proving ground shall use a soft impact area free from rocky debris so as to assure that the cartridge cases (projectiles) fired for recovery and examination are not deformed or become separated upon impact.

6.5 Proving ground test summary

	Sample Size				
Phase	First Article	Lot Acceptance	Requirement	Sampling Para.	Test Para.
70°±2.5°F	25	10	Safety	4.4.3.3	4.5.4

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project Number: 1315-A460

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63318 CASE CARTRIDGE 165MM M104 METAL PARTS ASSEMBLY

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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